

PAPER_246: MUGE Dual-Mode Oscillatory Gravity – Standing Wave and Hubble-Normalised Traveling Wave

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 – Star-Magic Physics **Source:** CondensedPhysics3.py – MUGEDualModeOscillatoryGravityCalculator (Session 62, grok_share_8d951e12.txt 4th-pass) **Date:** March 2026 **Series:** Phase 2 Session 62 – 3.x Universal MUGE Sub-Term Integration

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{UQFF}(r) = g_{MUGE}(r) \cdot \left(1 - [SSq] \cdot U_{bi} / F_U(r, t)\right), \quad [SSq] = 0.57$$

Abstract

Gravity in the MUGE framework is not a static field – it supports oscillatory modes that arise from the interference of inward- and outward-propagating gravitational perturbations. This paper establishes the **dual-mode oscillatory gravity sub-term** (`g_osc`), which is the superposition of two distinct wave modes: a standing wave and a Hubble-normalised traveling wave.

Mode 1 (standing wave): `g_osc1 = 2A*cos(kx)*cos(?t)` – the classic interference pattern of two equal-amplitude counter-propagating waves. Mode 2 (traveling wave): `g_osc2 = (2p/T_H_gyr)*A*cos(kx - ?t)` – a unidirectional propagating disturbance whose amplitude is suppressed by the inverse Hubble time in gigayears, `(2p/T_H_gyr)`, connecting gravitational oscillations to the cosmological expansion rate.

The key resonance condition – `?_local = 2p/t_Hubble` – places the system at the threshold where Mode 2 dominates the superposition because `(2p/T_H_gyr) ? 1` for `T_H_gyr = 2p`. Away from resonance, Mode 1 dominates for Hubble times much larger than 2p Gyr. The time-averaged result `?g_osc? = 0` ensures no net secular drift – oscillatory gravity is a zero-mean perturbation to the static MUGE field.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^4 \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis – establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. System Parameters and Equation Overview

Parameter	Symbol	Default Value	Units	Meaning
Oscillation amplitude	<code>A_osc</code>	1×10^{10}	m/s^2	Gravitational wave amplitude
Wavenumber	<code>k</code>	$1/r$	$1/\text{m}$	Spatial frequency at system scale <code>r</code>
Angular frequency	<code>?</code>	$2p/r$	rad/s	Relativistic frequency at scale <code>r</code>
Hubble time	<code>t_H_gyr</code>	13.8	Gyr	Current epoch value
Position	<code>x</code>	variable	m	Spatial coordinate

Parameter	Symbol	Default Value	Units	Meaning
Time	t	variable	s	Note: in context, passed as epoch

Primary equations:

Mode 1 (standing wave):

$$g_{osc1} = 2 * A * \cos(k*x) * \cos(?*t)$$

Mode 2 (Hubble-normalised traveling wave):

$$g_{osc2} = (2p / T_H_gyr) * A * \cos(k*x - ?*t)$$

Total:

$$g_{osc} = g_{osc1} + g_{osc2}$$

Time average:

$$\langle g_{osc} \rangle = 0 \quad (\text{both modes are zero-mean over integer wave periods})$$

Mode 2 amplitude factor at $T_H_gyr = 13.8$:

$$(2p / 13.8) \approx 0.455$$

2. Core Physics Derivation

2.1 Standing Wave – Counter-Propagating Superposition

A standing gravitational wave arises when two plane waves with equal amplitude A , wavenumber k , and frequency ω travel in opposite directions along x :

$$g_f = A * \cos(kx - \omega t) \quad [\text{forward}]$$

$$g_b = A * \cos(kx + \omega t) \quad [\text{backward}]$$

$$g_{standing} = g_f + g_b = 2A * \cos(kx) * \cos(\omega t) \quad [\text{trig identity}]$$

This mode is spatially modulated by $\cos(kx)$ – nodes at $kx = (n+\frac{1}{2})p$, antinodes at $kx = np$. At nodes, gravity is unaffected by the standing wave; at antinodes, the oscillation reaches full amplitude $2A$.

2.2 Traveling Wave – Hubble-Time Amplitude Suppression

Mode 2 is a single traveling wave whose amplitude is modulated by a cosmological suppression factor:

$$g_{osc2} = (2p / T_H_gyr) * A * \cos(k*x - \omega*t)$$

The factor $(2p / T_H_gyr)$ has units of $1/\text{Gyr}$ when T_H_gyr is in gigayears, but since A is already in m/s^2 , the result is dimensionally consistent only if T_H_gyr is treated as a dimensionless ratio $T_H / (1 \text{ Gyr})$. This convention is standard in cosmological normalisation within MUGE.

Physical interpretation: Gravitational disturbances traveling at cosmological speeds are attenuated by the Hubble expansion. The factor $(2p/T_H_gyr)$ is the instantaneous angular expansion rate in units of $(\text{Gyr})^{-1}$, analogous to the Hubble parameter $H_0 = 1/t_Hubble$ but expressed in the natural angular frequency unit.

2.3 Resonance Condition

At resonance, the traveling-wave amplitude equals the standing-wave amplitude:

$$(2p / T_H_gyr) = 1 \quad \Rightarrow \quad T_H_gyr = 2p \approx 6.28 \text{ Gyr}$$

For the current Universe ($T_{\text{H_gyr}} = 13.8$), Mode 2 amplitude factor 0.455 – Mode 2 carries about 45% of Mode 1 amplitude. At early cosmic times ($T_{\text{H_gyr}} \approx 6.3 \text{ Gyr}$, i.e., $z \approx 0.5$ in Λ CDM), the two modes were equal in amplitude. At Mode 2 resonance ($T_{\text{H_gyr}} = 2p$), the interference pattern is maximally complex.

2.4 Zero Time Average

Both sinusoidal modes average to zero over a complete oscillation period:

$$\begin{aligned} \langle \cos(kx) \cos(\omega t) \rangle_t &= \cos(kx) * \langle \cos(\omega t) \rangle_t = 0 \\ \langle \cos(kx - \omega t) \rangle_t &= 0 \\ \langle g_{\text{osc}} \rangle_t &= 0 \end{aligned}$$

This result ensures that oscillatory gravity is a **perturbative zero-mean correction** to the static MUGE field – it modulates gravity on timescale $2p/\omega = r/c$ (light-crossing time of the system) but produces no secular drift in the total gravitational potential.

2.5 Wavenumber and Frequency at Astrophysical Scales

For a system of radius r , the natural wavenumber and frequency are $k = 1/r$ and $\omega = 2pc/r$. This choice ties the oscillation period to the light-crossing time:

$$T_{\text{osc}} = 2p/\omega = r/c$$

For $r = 1 \text{ kpc}$: $T_{\text{osc}} \approx 3.3 \text{ kyr}$ – much shorter than stellar evolution timescales, so the oscillation averages out over physical processes. For $r = 1 \text{ Mpc}$: $T_{\text{osc}} \approx 3.3 \text{ Myr}$ – comparable to galaxy cluster merger timescales.

3. Dual-Mode Zero-Mean Theorem

Theorem (MUGE Oscillatory Zero Mean): The dual-mode oscillatory sub-term $g_{\text{osc}} = g_{\text{osc1}} + g_{\text{osc2}}$ is a zero-mean bounded perturbation to the static MUGE field for all systems with finite r . The maximum instantaneous amplitude is $|g_{\text{osc}}|_{\text{max}} = A * (2 + 2p/T_{\text{H_gyr}})$, reached when both modes constructively interfere at an antinode. No secular modification of total MUGE gravity results from this term in the time-averaged limit.

The **Hubble resonance condition** $T_{\text{H_gyr}} = 2p$ is the unique epoch at which Mode 1 and Mode 2 have equal amplitude, producing the most complex gravitational interference pattern observable.

4. Observational Predictions / Validation

- **Gravitational wave background:** The dual-mode structure predicts a specific spatial correlation pattern in the stochastic gravitational wave background – standing-wave nodes should appear as directions of suppressed GW strain in future pulsar timing arrays (IPTA, SKA).
 - **Galaxy cluster mass oscillations:** At $r \sim \text{Mpc}$, $T_{\text{osc}} \sim 3 \text{ Myr}$ – oscillatory gravity contributes to ICM pressure waves seen in Chandra X-ray maps. The mode-2/mode-1 amplitude ratio (0.455 at $z=0$) is a direct probe of the Hubble constant at the cluster.
 - **Early Universe enhancement:** At $z \approx 0.5$ ($T_{\text{H_gyr}} \approx 2p$), the standing and traveling waves were equal – enhanced gravitational perturbations at this epoch may leave an imprint in the large-scale galaxy power spectrum at $k \approx 0.1 \text{ h/Mpc}$.
-

5. References

1. Maggiore, M. (2007). *Gravitational Waves: Theory and Experiments*. Oxford University Press.
2. Riles, K. (2023). Gravitational waves: Sources, detectors and searches. *Prog. Part. Nucl. Phys.* 68, 1.
3. Planck Collaboration (2020). Planck 2018 Results I. *A&A* 641, A1.
4. Murphy, D.T. (2025). UQFF Framework v4.x – MUGE Sub-Term Integration. Star-Magic internal document.
5. grok_share_8d951e12 validation session – dual-mode oscillatory gravity term implementation.

PAPER_246 / UQFF v4.27 / Star-Magic / Session 62 / March 2026